BITS F464 Machine Learning, Semester I 2026–27

Assignment 1: Flat vs. Hierarchical Classification From Scratch

Results matrix

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Anuran Calls (MFCCs), with the 22 MFCCs as the only features. The handout's `stratified_split` (seed 42) gives 5756 training and 1439 test rows, and every model uses the settings of the handout's Table 1. Accuracies are in %. Every number in Tables 2 and 3 is printed by `part1_decision_tree.py` to `part5_naive_bayes.py` (Python 3.12.10, NumPy 2.5.4, pandas 3.0.6).

Table 2: Results matrix (flat models predict Species; † see the note).

		Overall accuracy (Species)		Family acc. †
Method	Criterion	Train	Test	Test
1. Single Decision Tree	Gini	100.00	94.93	96.25
	Entropy	100.00	95.41	96.80
2a. Bagging	Gini	99.97	96.66	97.50
	Entropy	99.98	96.87	97.50
2b. AdaBoost	Gini	97.52	95.34	96.73
	Entropy	96.35	94.16	95.00
3. Random Forest	Gini	100.00	97.78	98.40
	Entropy	100.00	97.64	98.19
4. Hierarchical Decision Tree	Gini	100.00	93.82	**Router:** 95.07
	Entropy	100.00	94.65	**Router:** 95.62
5. Gaussian Naive Bayes	—	91.47	90.34	91.94

† For flat models, the *derived* family accuracy: each predicted species mapped to its family. For the hierarchical rows, the router accuracy; their overall accuracy columns hold the final species accuracy.

Table 3: Supplementary: hierarchical failure analysis (test set, %).

Criterion	Router acc.	Oracle acc.	(A) Router right, expert wrong	(B) Router wrong
Gini	95.07	98.26	1.25	4.93
Entropy	95.62	98.33	0.97	4.38

$A_{\text{overall}} = A_{\text{router}} - p_A$, for example $95.07 - 1.25 = 93.82$ for Gini; `part4_hierarchical.py` checks it exactly, on counts.

What the numbers show

Pairs of numbers a / b are Gini / entropy; accuracies are on the test set unless stated otherwise. 55 of the 60 recordings have syllables on both sides of the split, so every accuracy here is optimistic for new recordings.

Single tree (Part 1). Grown without limits, both trees fit the training set exactly (100%; depth 17 / 11, with 211 / 181 leaves) but reach only 94.93 / 95.41% on the test set: a full tree has low bias and high variance. Table 4 shows the trade-off. Shallow trees underfit both sets; test accuracy levels off by depth 12 while training accuracy goes on to 100%, so the deepest splits fit noise in the training rows, not structure. Entropy is ahead of Gini at every depth on this data. The derived family accuracy (96.25 / 96.80%) is higher than the species accuracy because a species error inside the right family still counts as a family hit.

Table 4: Single tree against `max_depth` (not part of the matrix), %.

		`max_depth`						
Criterion	Set	1	2	3	5	8	12	None
Gini	Train	60.98	69.13	78.51	90.10	97.85	99.81	100.00
	Test	61.01	69.28	79.22	88.60	93.95	94.86	94.93
Entropy	Train	61.78	73.38	84.40	91.80	98.68	100.00	100.00
	Test	62.40	73.94	84.50	90.90	94.86	95.41	95.41

Bagging and AdaBoost (Part 2). Bagging 25 trees gains 1.73 / 1.46 points over the single tree. Each tree overfits its own bootstrap sample differently, and the vote averages those errors away (variance reduction); training accuracy stays at 99.97 / 99.98%, since a row that only a minority of the trees saw can still be outvoted. AdaBoost attacks bias instead: one depth-3 tree reaches 79.22 / 84.50% (Table 4), and 50 reweighted rounds lift that to 95.34 / 94.16%, 0.41 / −1.25 points against the single unconstrained tree. All 50 rounds were kept: the weighted error grows from 0.21 / 0.16 to 0.60 / 0.49 as the weights move onto the hard rows, but stays below the limit $1 - 1/K = 0.9$. Decision stumps reach only 83.32 / 84.16%. A stump has two leaves, so it names at most two of the ten species per round; its weighted error climbs to 0.78 / 0.80, close to the limit, which leaves each late stump a small say ($\alpha = 0.93$ / 0.82, against 1.77 / 2.25 for the last depth-3 trees), and the 50-stump vote predicts only 8 / 9 of the ten species on the test set.

Random forest (Part 3). The forest is the best flat model: 97.78 / 97.64%, 1.12 / 0.77 points above bagging and 2.85 / 2.23 above the single tree. Its trees are individually weaker than bagged trees (mean single-tree accuracy 92.94 / 93.07% against 93.67 / 94.31%), because each split sees only 4 of the 22 MFCCs, but they agree less often: the average pairwise agreement of their test predictions is 0.899 / 0.903, against 0.922 / 0.932 for bagging. Less correlated errors cancel in the vote, so the ensemble gains more than its members lose.

Hierarchy (Part 4). The hierarchy is 1.11 / 0.76 points below the flat tree (93.82 / 94.65% against 94.93 / 95.41%). The experts are not the problem: given the true family they reach 98.26 / 98.33% (the oracle), and only 1.25 / 0.97% of the rows are lost by an expert after correct routing. The router is the bottleneck. It sends 4.93 / 4.38% of the rows to the wrong family, each then a certain species error, and its accuracy (95.07 / 95.62%) is 1.18 / 1.18 points below the family accuracy that the flat tree gets for free by mapping its predicted species, 17 / 17 test rows: Hylidae 9 / 3, Leptodactylidae 9 / 9, Dendrobatidae 0 / 4 and Bufonidae −1 / 1. A likely reason: each family is a union of species clusters (five in Hylidae, three in Leptodactylidae), which one tree must carve out from the coarse labels alone. Router recall is lowest on the small families (Bufonidae, 14 test rows: 71.43 / 71.43%), where the experts are constants and routing is everything, but the flat tree is weak there too (64.29 / 78.57%). A perfect router would add 4.44 / 3.68 points. On this dataset the easier sub-problems do not make up for the router's mistakes, so the hierarchy does not help.

Naive Bayes (Part 5). Naive Bayes is the weakest model in the matrix (90.34%). It treats the 22 MFCCs as independent within each species, but they are not: within a species the mean $|r|$ between two features is 0.23 to 0.50. Over the whole training set 43 of the 231 pairs have $|r| > 0.5$, up to 0.91, though that also reflects the differences between the species. Correlated features count the same evidence several times, so a block of related MFCCs can outvote the rest; a tree uses one feature per split and is not misled by redundancy. A second effect is specific to this data: MFCC 1 is constant in every training row of Ameeregatrivittata, so its variance there is only $\varepsilon = 7.03 \times 10^{-11}$. Yet 5563 of the 5756 training rows share that value and gain a large likelihood bonus: naive Bayes predicts the species for 187 test rows, 79 of them wrongly.